

PROJECT SYNOPSIS

Title

Patient-Friendly Discharge Summary Generator Using LLM and Retrieval-Augmented Generation (RAG)

Introduction

Healthcare institutions generate discharge summaries that contain important information regarding a patient's treatment, medication changes, laboratory findings, follow-up instructions, and warning signs. However, these summaries are often written in complex medical terminology that is difficult for patients to understand. Poor comprehension of discharge instructions can lead to medication errors, missed follow-up appointments, and increased hospital readmissions.

This project proposes an AI-powered Discharge Summary Generator that utilizes Large Language Models (LLMs) combined with Retrieval-Augmented Generation (RAG) to automatically create patient-friendly discharge summaries from Electronic Health Records (EHRs). The system converts clinical information into simple, understandable language while preserving medical accuracy.

Problem Statement

Develop a discharge summary generator that compiles medication changes, follow-up instructions, and warning signs in simple language.

Objectives

1. To extract relevant patient information from EHR notes, medication records, and laboratory reports.
2. To employ RAG techniques for retrieving accurate medical context.
3. To generate discharge summaries in patient-friendly language using an LLM.
4. To reduce medical jargon and improve patient understanding.

Proposed System

The proposed system accepts patient data from Electronic Health Records, including clinical notes, medication lists, diagnosis details, and laboratory results. A Retrieval-Augmented Generation framework retrieves relevant medical knowledge and contextual information from a healthcare knowledge base.

The retrieved information is combined with patient-specific data and provided to a Large Language Model. The model generates a simplified discharge summary containing Patient diagnosis overview, Medication changes and dosage instructions, Follow-up appointment recommendations, Lifestyle and recovery guidelines, important warning signs requiring medical attention, Contact information and emergency instructions. The generated summary is presented in clear, non-technical language suitable for patients and caregivers.

Methodology

Step 1: Data Acquisition

Collect patient information from EHR records, medication databases, and laboratory reports.

Step 2: Preprocessing

Extract relevant clinical entities such as diagnosis, medications, test results and physician recommendations.

Step 3: Knowledge Retrieval

Store medical guidelines and discharge protocols in a vector database. Retrieve relevant context using semantic search.

Step 4: Summary Generation

Use an LLM to combine patient data and retrieved knowledge to generate a comprehensive discharge summary.

Step 5: Validation

Evaluate readability, completeness, and medical accuracy through testing and expert review.

Technologies Used

Programming Language: Python

Framework: LangChain / LlamaIndex

LLM: GPT, Llama, or Mistral

Vector Database: FAISS, ChromaDB, or Pinecone

Database: PostgreSQL / MongoDB

PDF Generation: ReportLab

Input

EHR Notes, Medication List, Laboratory Results and Physician Recommendations

Output

Simplified Patient Discharge Summary, Medication Instructions, Follow-up Schedule and Warning Signs and Emergency Guidance

Expected Outcomes

- ✓ Improved patient understanding of discharge instructions.
- ✓ Reduction in medication-related misunderstandings.
- ✓ Better adherence to follow-up care.
- ✓ Lower risk of hospital readmissions.
- ✓ Enhanced healthcare communication.

Conclusion

The Patient-Friendly Discharge Summary Generator leverages Large Language Models and Retrieval-Augmented Generation to bridge the communication gap between healthcare providers and patients. By transforming complex medical information into simple, understandable language, the system improves patient engagement, treatment adherence, and overall healthcare outcomes while reducing the burden on medical professionals.